Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Two series components have drops 3 V and 5 V. A charge of 7 C passes through each. Find each energy transfer and the total.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$.
Series resistors share one unbranched current path.

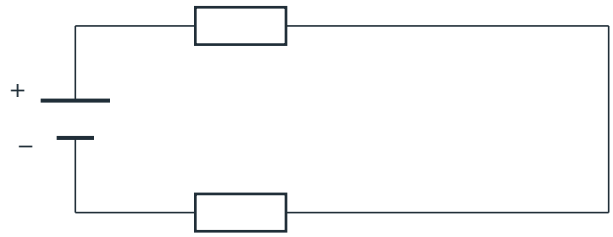
Worked reasoning: Two series resistors 2 and 4 ohms on a 12 V ideal supply have total 6 ohms and current 2 A. Drops are 4 V and 8 V; they add to 12 V.

Your turn: Draw the loop and label current and both voltage drops.

Marzano check - circle: ① ② ③

03 Represent and interpret

Trace the only current path in the series diagram. If either resistor connection opens, what happens to the current?



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

A 12 V battery powers 10 Ω and 20 Ω series resistors. Find total resistance, current and each voltage drop.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student assigns more current to the smaller series resistor. Explain why this would imply charge accumulating between resistors.

Marzano check - circle: 1 2 3

06 Choose a method

An object moves 0.4 m toward a fixed plane mirror. Find image movement and the change in object-image separation.

Marzano check - circle: 1 2 3

07 Exit check

Add a third equal resistor to two equal series resistors at the same supply voltage. Compare new current with original current.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Three series resistors 2, 3, R ohms draw 2 A from 20 V. Find R and power in each resistor.

Marzano check - circle: 1 2 3